FIT3181/5215 Deep Learning

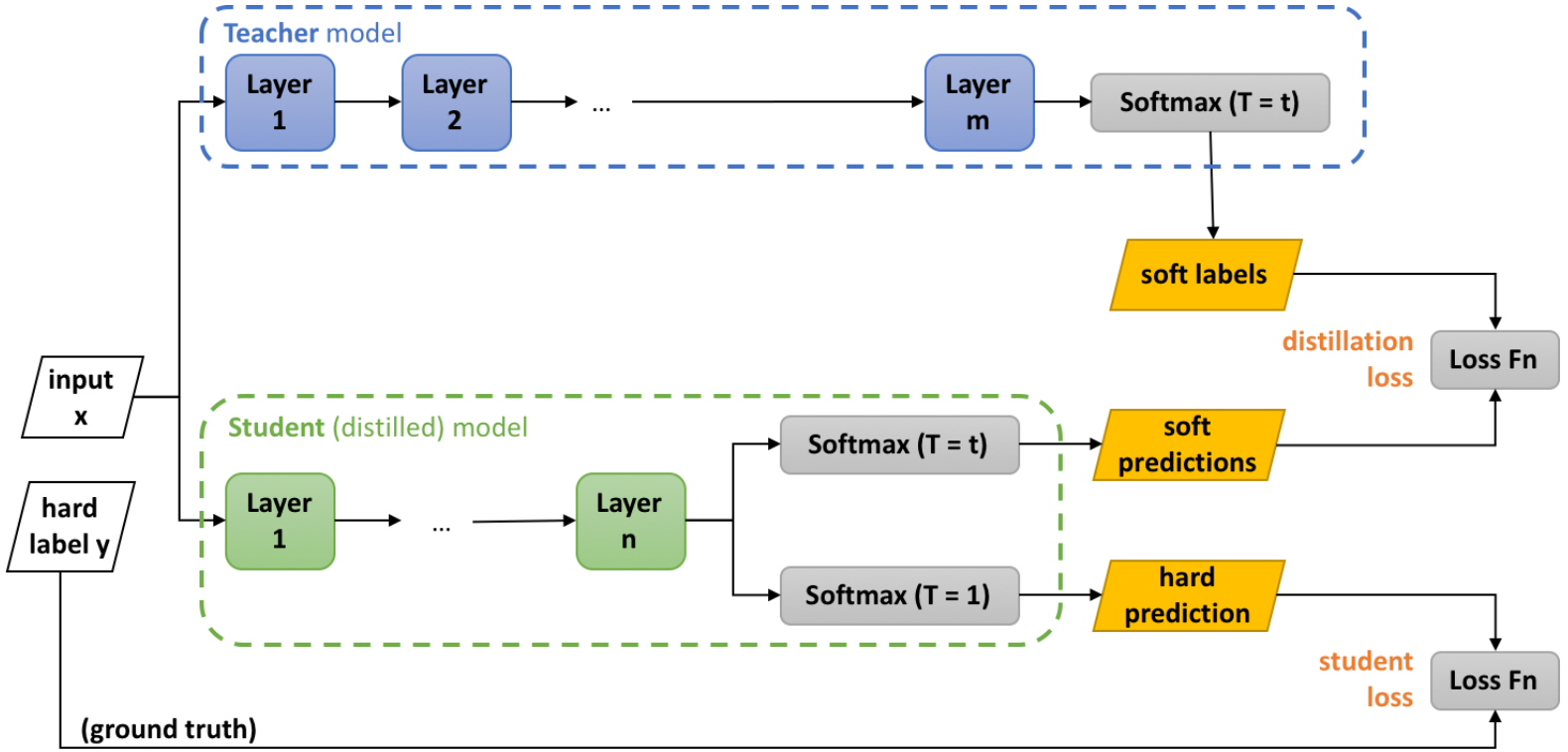
Knowledge Distillation

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Knowledge Distillation

1. **What:** transfer knowledge from a complex model (Teacher) to a simpler model (Student)
2. **Why:** reduce computational complexity
3. **How:** train a compact deep neural networks to reproduce the output or mimic the behavior of a large complex model (which can be pre-trained or trained concurrently with the student)



Knowledge Distillation

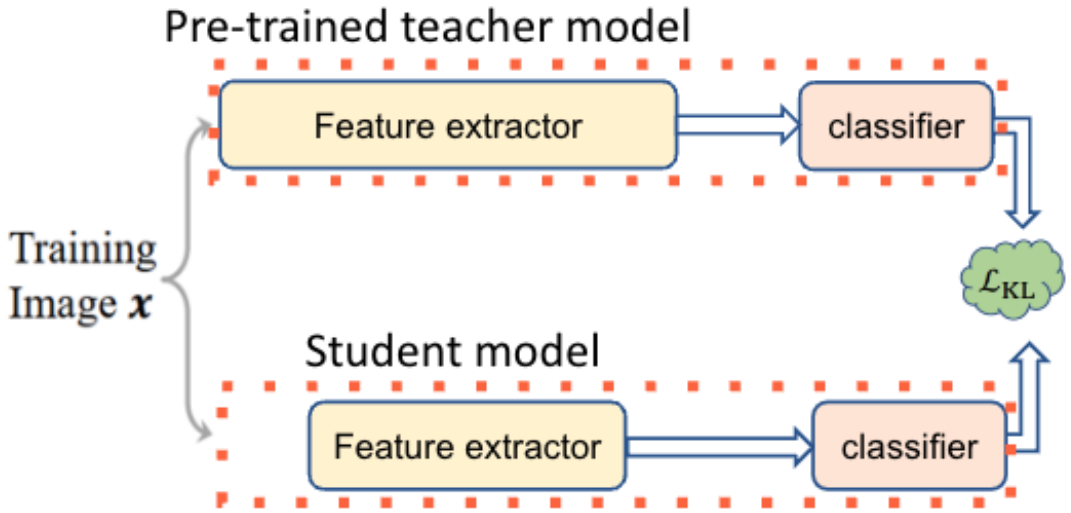
There are two main families of distillation techniques

Output distillation:

- Cross-entropy (KD) [1]

Feature distillation: mimic the hidden features of teacher network

- Feature response (FITNET) [2]
- Spatial attention (AT) [3]
- Feature distribution (PKT) [4]



Output distillation



Feature distillation: Attention transfer [3]

[1] Geoffrey Hinton et. al. Distilling the Knowledge in a Neural Network. In NIPS Workshop, 2014
[2] Adriana Romero et. al. Fitnets: Hints for thin deep nets. In ICLR, 2015.
[3] Nikos Komodakis and Sergey Zagoruyko. Paying more attention to attention: improving the performance of convolutional neural networks via attention transfer. In ICLR, 2017.
[4] Nikolaos Passalis and Anastasios Tefas. Learning deep representations with probabilistic knowledge transfer. In ECCV, 2018

Resources

1. Geoffrey Hinton et. al. **Distilling the Knowledge in a Neural Network.** In NIPS Workshop, 2014. https://arxiv.org/abs/1503.02531v1 - 2015

2. Adriana Romero et. al. **Fitnets: Hints for thin deep nets.** In ICLR, 2015. https://arxiv.org/abs/1412.6550

3. Nikos Komodakis and Sergey Zagoruyko**. Paying more attention to attention: improving the performance of convolutional neural networks via attention transfer.** In ICLR, 2017. https://arxiv.org/abs/1612.03928

4. Nikolaos Passalis and Anastasios Tefas. **Learning deep representations with probabilistic knowledge transfer.** In ECCV, 2018. https://arxiv.org/abs/1803.10837

Resources (Advanced)

1. Yuan, Li, et al. **Revisiting knowledge distillation via label smoothing regularization.** CVPR. 2020. https://arxiv.org/abs/1909.11723

2. Zhao, Borui, et al. **Decoupled knowledge distillation.** CVPR. 2022. https://arxiv.org/abs/2203.08679

3. Chen, Pengguang, et al. **Distilling knowledge via knowledge review.** CVPR. 2021. https://arxiv.org/abs/1503.02531

Thanks for your attention!